Operation Instructions

These are the operation instructions for the DIY LED Clock firmware found here:

https://github.com/fabianswebworld/stc-led-clock-firmware

1. Buttons and indicators

The clock is operated using two push-buttons located on the side or rear of the unit:

- **Button 1 (S1, top):** Main menu / Advance menu item / Select value
- **Button 2 (S2, bottom):** Next display page / Select menu item / Increment value

The dots on the display are used as indicators:

- **Center Colon (`:`):** Flashes once per second with an incandescent filament effect (gradual ignition and afterglow decay).
- **Indicator Dot:**
  - In 24-hour display mode: The decimal point at the lower-right corner of the fourth digit stays steadily lit when the daily alarm is armed.
  - In 12-hour display mode: The decimal point at the lower-right corner of the fourth digit indicates PM.

In normal operation, the clock displays the current hours and minutes. Additionally, at configurable intervals, an "auto-rotation" through other items (Temperature, Date, Day of Week) is performed. For each of these three items (Temperature, Date, Day of Week), the inclusion in the auto-rotation can be enabled or disabled.

2. Actions from normal time display

- **Button 1 (Short press):** Enters the Main menu (see Section 7).
- **Button 1 (Hold for 1 second):** Launches the 7-segment Snake animation manually.
- **Button 2 (Short press):** Steps to the next display page (see Section 3).
- **Button 2 (Hold for 1 second):** Enters the Countdown timer (see Section 4).

3. Manual display of other information

Pressing **Button 2** steps through the available information displays in sequence:

```
Time -> Temperature -> Date -> Day of Week -> Year-> Seconds -> Time
```

- **Temperature:** Displays ambient temperature (e.g., `22°C` or `72°F`).
- **Date:** Displays month and day according to the selected format (e.g., `31.12.`, `31.12`, or `12.31`).
- **Day of Week:** Displays three-letter abbreviation of the current weekday.
- **Year Display (`'xx`):** Displays the last two digits of the year preceded by an apostrophe at the top (e.g., `'26` for 2026). The internal clock circuit automatically calculates leap years.
- **Seconds Display (`MM.SS`):** Displays elapsed minutes on the left digits and running seconds on the right digits.

- **Time Synchronization:** Pressing **Button 1** while on this page immediately resets the seconds counter to `.00` to synchronize with a reference clock.

When manually advancing the modes, all of the above pages are displayed, regardless of whether they are also enabled for the auto-rotation in the `dISP` menu (see Section 5). A page selected manually with Button 2 stays on the screen indefinitely, the clock will not switch back automatically.

- **Returning to Clock:** Pressing **Button 2** until the hours and minutes return restores normal timekeeping and re-enables automatic rotation.

4. Countdown Timer (`Ctdn`)

1. **Activation:** While in normal time display, **press and hold Button 2**. The display shows `Ctdn`.
    - Press **Button 1**: Exits back to the clock without starting a timer.
    - Press **Button 2**: Enters timer setting mode.
2. **Setting the Duration (`MM:SS`):**
    - The display shows the last used timer preset (factory default: `00:59`). The colon remains steady.
    - **Button 1:** Increments minutes
    - **Button 2:** Increments seconds
    - **Button 1 + Button 2 (simultaneously):** Resets the timer to `00:00`.
    - **Hold Button 2:** Cancels countdown and returns to normal time.
3. **Automatic Start:**
    - Leave buttons untouched for **2 seconds** - the countdown begins automatically (the starting value is saved to memory as the new default preset).
    - The colon flashes at 1 Hz while counting down.
4. **While running:**
    - Pressing any button while running immediately pauses the countdown and goes back to the Duration setting, allowing to quickly add more time, starting from the current value.
    - **Hold Button 2:** Cancels countdown and returns to normal time.
5. **Timer Alarm:**
    - When reaching `00:00`, the display flashes rapidly and the buzzer sounds.
    - **Pressing either Button 1 or Button 2** silences the alarm and immediately returns to the normal clock display.

5. Automatic Display Rotation

If enabled in the `diSP` menu (see Section 7), the clock will smoothly dissolve from the time display through the enabled information pages (Temperature, Date, Day of Week) before returning to the clock. This rotation can be configured in the `dISP` menu (see Section 7):

- **Cadence/Interval Options (`rot`):**
  - `rot0`: Automatic rotation disabled (manual cycling only).
  - `rot1`: Once per minute (at second :30).
  - `rot2`: Twice per minute (at seconds :00 and :30).
  - `rot3`: Three times per minute (at seconds :00, :20, and :40).
- **Page Duration (`dur`):**
  - Configurable from **`dur1` to `dur5`** (each page holds for 1 to 5 seconds).

6. Hourly animation (`Ani`)

An fun little snake animation starts: the snake slithers across the display segments, locates "apples", grows in length, and dissolves back into the clock display.

- **Cadence Options (`Ani`):**
  - `Ani0`: Animation disabled.
  - `Ani1`: Twice daily (at 12:00 noon and 00:00 midnight).
  - `Ani2`: Once every hour (at the top of the hour, :00), default.
  - `Ani3`: Every half hour (:00 and :30).
  - `Ani4`: Every quarter hour (:00, :15, :30, :45).
  - `Ani5`: Every minute (at second :00).
- **Manual Trigger:** Holding **Button 1** for 1 second in normal clock mode launches the animation immediately.

7. Settings Menu

Press **Button 1** from normal clock mode to enter the main menu tree.

- Use **Button 1** to cycle through the four top-level menus (`SEt`, `bEEP`, `diSP`, `ConF`).
- Use **Button 2** to enter a menu.
- Inside any setting: **Button 2** changes the value, **Button 1** confirms and advances to the next setting.

```
[Normal Clock]
     |
  (Button 1)
     |
     +---> [ SEt  ] ---> Clock Time -> Alarm Time -> Chime Hours (begin/end) -> Date -> Day of Week
     |                   -> Year
     |
     +---> [ bEEP ] ---> Alarm Toggle -> Hourly Chime Toggle
     |
     +---> [ diSP ] ---> Temperature Enable -> Date Enable -> Day of Week Enable -> Rotation cadence
     |                   -> Page duration -> Animation cadence
     |
     +---> [ ConF ] ---> 12/24h -> Leading Zero -> Temperature  Unit -> Date Format -> Brightness
                         -> Temperature Calibration -> Firmware Version
```

Menu 1: Set Clock, Alarm, Date (`SEt`)

Press **Button 2** on `SEt` to enter the setting sequence:

1. **Clock Time (`CL`):** Hours flash -> press Button 2 to adjust -> press Button 1 -> Minutes flash -> press Button 2 to adjust -> press Button 1.

2. **Daily Alarm Time (AL):** Alarm hours flash -> adjust with Button 2 -> press Button 1 -> Alarm minutes flash -> adjust with Button 2 -> press Button 1.
3. **Hourly Chime Window (CH):** Sets active hours for the chime.
    - Start Hour flashes -> adjust with Button 2 -> press Button 1.
    - End Hour flashes -> adjust with Button 2 -> press Button 1.
4. **Date (DA):** Month flashes -> adjust with Button 2 -> press Button 1 -> Day flashes -> adjust with Button 2 -> press Button 1.
5. **Day of Week (DY):** Day flashes -> press Button 2 to adjust (Sun ... Mon) -> press Button 1.
6. **Year (Yr):** Year flashes -> press Button 2 to adjust (00 ... 99) -> press Button 1 to save and exit to clock.

Menu 2: Sound Settings (bEEP)

Press **Button 2** on bEEP to enter:

1. **Alarm Enable:** Displays AL on or AL oF. Press Button 2 to toggle. Press Button 1 to confirm.
2. **Chime Enable:** Displays CH on or CH oF. Press Button 2 to toggle. Press Button 1 to confirm and exit to clock.

Menu 3: Display Settings (diSP)

Press **Button 2** on diSP to enter settings for the display rotation:

1. **Temperature Page:** Displays tP on or tP oF. Press Button 2 to toggle. Press Button 1 to advance.
2. **Date Page:** Displays dA on or dA oF. Press Button 2 to toggle. Press Button 1 to advance.
3. **Day of Week Page:** Displays dy on or dy oF. Press Button 2 to toggle. Press Button 1 to advance.
4. **Auto-Rotation Cadence (rot):** Displays rot0, rot1, rot2, or rot3 (see Section 5). Press Button 2 to cycle. Press Button 1 to advance.
5. **Page Duration (dur):** Displays dur1 to dur5 (seconds to show each page in rotation). Press Button 2 to cycle. Press Button 1 to advance.
6. **Animation Cadence (Ani):** Displays Ani0 to Ani5 (see Section 6). Press Button 2 to cycle. Press Button 1 to save and exit to clock.

Menu 4: Configuration (ConF)

Press **Button 2** on ConF to enter Configuration menu:

1. **Time Format (12.24):** Press Button 2 to toggle between 12 (12-hour AM/PM) and 24 (24-hour). Press Button 1 to advance.
2. **Leading Zero Suppression (ZEro):** Press Button 2 to toggle between 09:59 (leading zero visible) and 9:59 (leading zero blanked). Press Button 1 to confirm and advance.

3. **Temperature Unit (`Unit`):** Press Button 2 to toggle between `°C` (Celsius) and `°F` (Fahrenheit). Press Button 1 to advance.
4. **Date Format (`DATE`):** Press Button 2 to cycle through:
    - `12.31` (Month.Day)
    - `31.12` (Day.Month)
    - `31.12.` (Day.Month with trailing point, German format)
    - Press Button 1 to confirm and advance.
5. **Automatic Dimming Range (`brt`):**
    - Press Button 2 to set maximum brightness level (01 ... 63).
    - Press Button 1, then Button 2 to set minimum (night-time) brightness level (01 ... 63).
    - Press Button 1 to advance.
6. **Temperature Sensor Calibration (`CALI`):**
    - Press Button 2 to enter -> current temperature reading is displayed.
    - Press Button 2 to increment offset (-15 ... +15 degrees).
    - Press Button 1 to advance.
7. **Firmware Version Display (`vEr`):**
    - Press Button 2 to enter -> display shows installed firmware revision (e.g. `1.2.0`).
    - Press Button 1 to exit to clock.

8. Backup Battery

The clock uses an onboard CR1220 3V coin cell battery to back up the DS1302 real-time clock circuit.

- **Retained Data:** Timekeeping, leap-year calculations, alarm settings, rotation preferences, countdown presets, and sensor calibrations are retained in non-volatile battery-backed memory.
- **Mains Reconnection:** When main power is disconnected and restored, the display immediately resumes operation with current time and customized settings intact without requiring reconfiguration.